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Consciousness came before life

The fundamental cause of evolution



8th May 2024

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Stuart Hameroff | Professor of Astrobiology, Psychology and Anesthesiology, and Director of the Center for Consciousness Studies at The University of Arizona, Tucson, Arizona.

Anirban Bandyopadhyay | Principal Research Scientist at the National Institute for Materials Science (NIMS) in Tsukuba, Japan.

Dante Laurotta | Regents Professor of Planetary Science and Cosmochemistry, and Director of the Arizona Astrobiology Center, The University of Arizona, Tucson, Arizona. He led NASA's OSIRIS REX mission to Asteroid Bennu.

3,022 words

Read time: approx. 15 mins

Most scientists believe that consciousness came after life, as a product of evolution. But observations of extraterrestrial organic material, along with Roger Penrose and Stuart Hameroff's quantum theory of consciousness, provide reason to believe that consciousness came before life. In fact, argue Hameroff and his collaborators, consciousness may have been what made evolution and life possible in the first place.

You can see Stuart Hameroff live, debating in 'Darwin vs Consciousness' alongside biologist Denis Noble and philosopher Antonella Tramacere at the upcoming [HowTheLightGetsIn Festival](#) on May 24th-27th in Hay-on-Wye.

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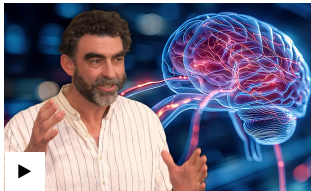
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Most scientists and philosophers believe that life came before consciousness. Life appeared on Earth about 3.8 billion years ago; consciousness and feelings, it's said, evolved later due to complex biological information processing, perhaps only recently in brains with language and tool-making abilities. In fact, though, there's good reason to think that consciousness preceded life, and was central to making life and evolution possible.



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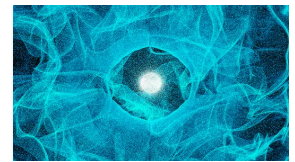
What is life? It is often described as its functions: metabolism, adaptation, reproduction, etc. But non-biological systems can have similar functions, for example, oceanic hydrothermal vents can metabolize, transform energy and synthesize chemicals, weather and climate systems adapt to changes in solar radiation, volcanic activity, and other natural factors, and a seed crystal in a solution can lead to the formation of more crystals with the same lattice structure, essentially reproducing itself. In the 19th century “[vitalists](#)” proposed life was a living field, force, or *élan vital*, but vitalism was eclipsed by molecular biology and genetics.

Erwin Schrödinger suggested that [a form of “quantum vitalism”](#) accounted for life’s unitary oneness, and Herbert Fröhlich later proposed that [quantum coherent vibrational modes](#), similar to those in a laser, could play a central role in various biological processes (“Fröhlich coherence”). The idea is that in a crystal-like structure, laser-like coherent vibrations could be driven by small random changes in temperature or energy.

Penrose argues that consciousness is made of collapses of quantum superpositions into definite states

This was proven by Anirban Bandyopadhyay’s group for biological microtubules – self-assembling cylindrical lattice polymers of the protein tubulin. Microtubules dynamically organize the interiors of all animal and plant cells, as part of the cell’s structural cytoskeleton, and appear also to serve as its nervous system and memory bank.

Anirban’s team sent low-power electromagnetic signals of varying frequency into microtubules and measured their conductance. They found distinctive self-similar



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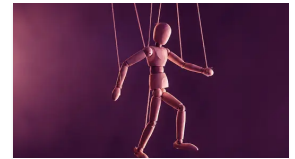
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resonance conductance patterns where each conductance response is grouped into sets of three, and then these groups are themselves grouped into larger sets of three. These “triplets-of-triplets” repeat in microtubules every three orders of magnitude, in kilohertz, megahertz, gigahertz and terahertz.

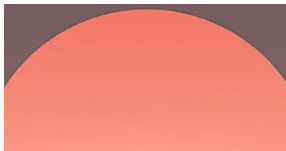
The “triplets-of-triplets” are phase shifted to make a unique structure called a “time crystal” (first described for biology by Arthur Winfree in the 1960s). In ordinary crystals, like salt or quartz, atoms are arranged in a repeating spatial pattern. But in a time crystal patterns also repeat in time, in a regular, repeating cycle returning to the same positions over and over. This movement happens without using energy – it’s a stable, perpetual motion at the atomic level.



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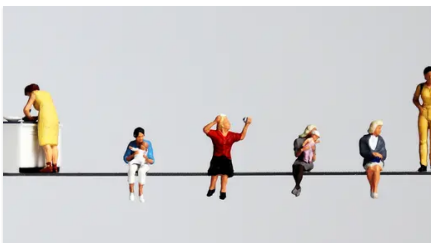
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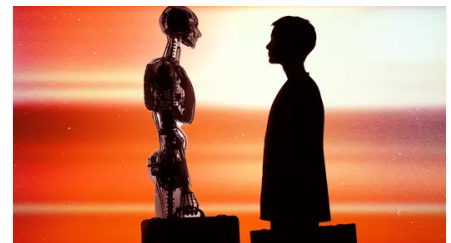
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